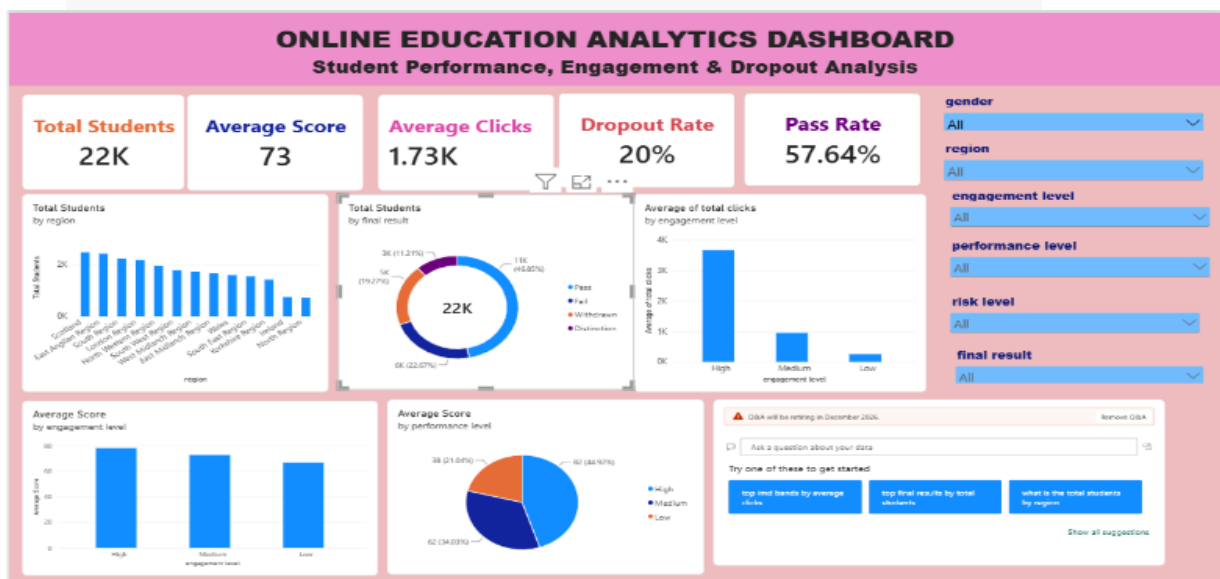
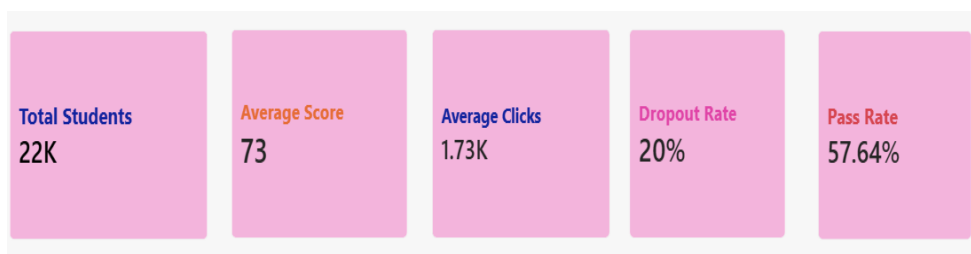


Report

Date	21 Sep 2026
Team ID	Master of Computer Applications
Project Name	Online Learning Engagement and Student Performance Analysis
Maximum Marks	5 Marks

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. It is typically used for in-depth analysis, documentation, and communication of results. Reports are suitable for a diverse audience, including decision-makers, analysts, and stakeholders who need a comprehensive understanding of the data.

Designing a report in Power BI involves connecting to data sources, creating visualizations like charts and graphs, customizing their appearance and interactivity, organizing them logically on the canvas, formatting elements for consistency and clarity, and optionally creating dashboards for a summarized view. Throughout the process, it's essential to consider the audience's needs and ensure the report effectively communicates insights from the data. Finally, iterate based on feedback to continually improve the report's design .



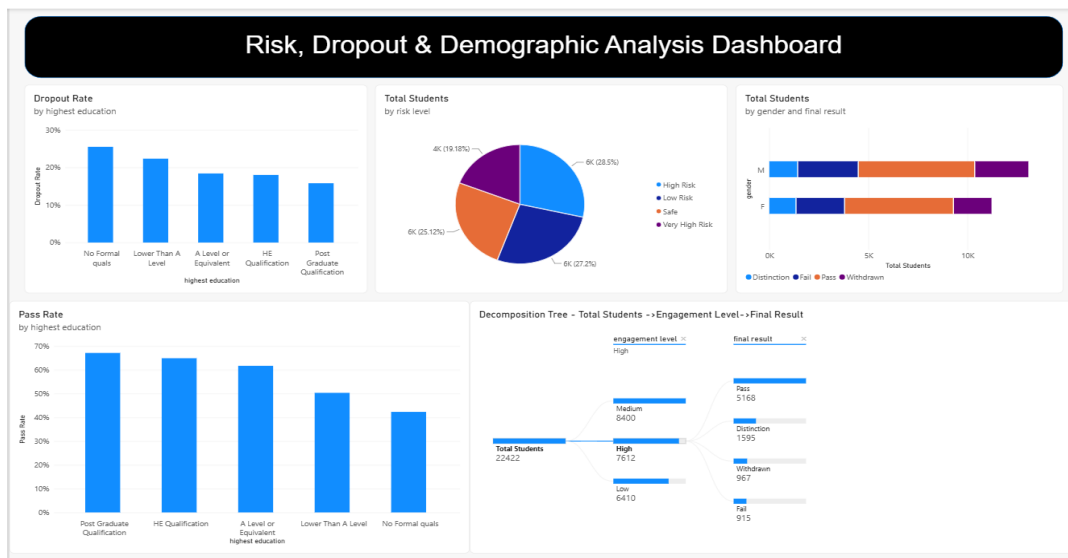
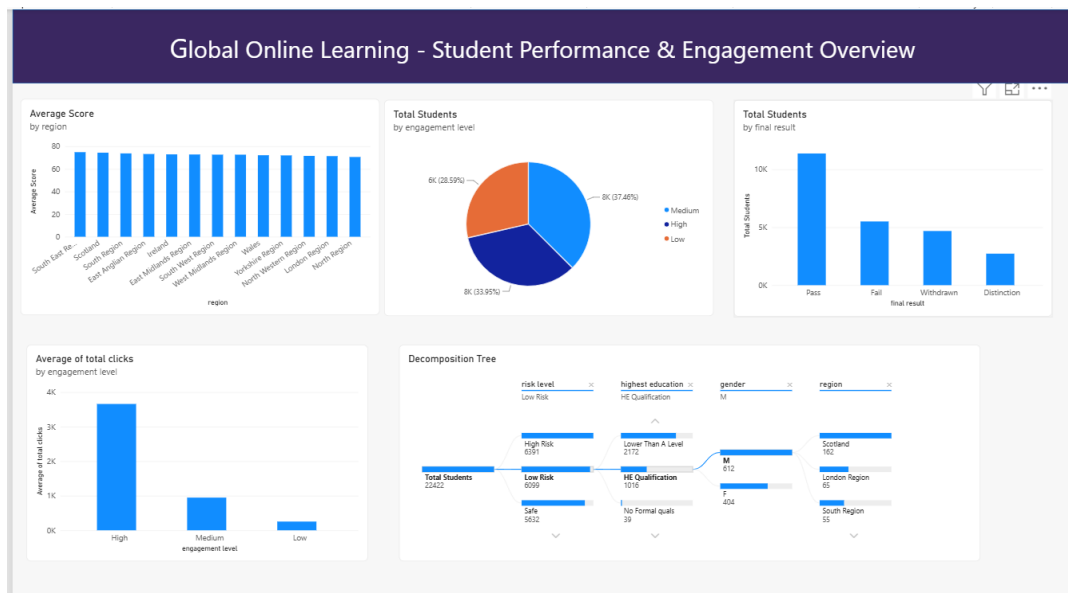


Figure: Layout of the four report pages (KPI Cards, Dashboard, Report 1, Report 2) built in Power BI on the online_education_dataset table.

Observations drawn from the Online Education Analytics report provide valuable insights into student performance, engagement and dropout trends across regions, education levels and risk groups.

1. Trends Over Time:

- The dataset is a student-level snapshot, so trends are read across behaviour bands rather than dates: average score and average total clicks rise together from Low to High engagement, showing how learning activity builds towards results.

- The click-band query (0–500, 500–1000, 1000–2000, 2000–3000, 3000+ clicks) shows whether scores keep improving with more activity or level off after a point.

2. Performance Comparisons:

- Average score, average clicks and student count are compared by **region** to show which regions lead and which lag (Page 3 – Average Score by Region; Page 2 – Total Students by Region).
- Highest-education groups are compared on both **Pass Rate** and **Dropout Rate** (Page 4) to see which prior-education levels succeed or struggle.

3. Student Segmentation:

- Students are segmented by engagement level, performance level, risk level, gender, region and highest education. The decomposition trees let a viewer drill from Total Students into Risk Level → Highest Education → Gender → Region.
- The Page 4 bar chart splits gender by final result (Pass / Fail / Withdrawn) so gender differences in outcomes are visible at a glance.

4. Goal Achievement:

- The five KPI cards – **Total Students**, **Average Score**, **Average Clicks**, **Dropout Rate** and **Pass Rate** – measure course health against targets. A high Pass Rate with a low Dropout Rate indicates the programme is meeting its goals.
- Dropout Rate by engagement level and by performance level identifies the segments where intervention is needed most.

Example:

1. KPI Overview: Score] and an average of [Average Clicks] clicks per student. The overall Pass Rate is [Pass Rate %] and the Dropout Rate is [Dropout Rate %].age Clicks

- The dashboard covers [Total Students] unique students with an average score of [Average

2. Engagement vs Performance:

- [High / Medium / Low] engagement students record the highest average score ([score]), while [Low] engagement students score only [score]. Higher click activity is associated with better results, so encouraging platform activity should improve outcomes.

3. Dropout Analysis:

- The **[engagement level]** group has the highest dropout rate at **[rate %]**, and the **[performance level]** group at **[rate %]**. Low engagement is therefore an early-warning signal for dropout.

4. Risk Distribution:

- The risk-level pie chart shows **[risk level]** as the largest group (**[count]** students, **[share %]**). Students in the high-risk group should be prioritised for support.

5. Regional Disparities:

- **[Region]** has the most students (**[count]**) while **[Region]** has the highest average score (**[score]**). **[Region]** trails on score and clicks, suggesting a need for regional support.

6. Education Level Analysis:

- Students with **[education level]** achieve the highest Pass Rate (**[rate %]**), while **[education level]** shows the highest Dropout Rate (**[rate %]**). Learners with lower prior education may need extra guidance.

7. Gender and Final Result:

- Of the **[count]** male and **[count]** female students, the final result split (Pass / Fail / Withdrawn) shows **[observation]**.

8. Recommendations:

- Target low-engagement and high-risk students with reminders and mentoring, add support for regions and education levels with high dropout, and track Pass Rate and Dropout Rate monthly against agreed goals.